Whole-Space Constrained Minimizers for a Two-Dimensional Fourth-Order Cubic–Quintic Hamiltonian

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Abstract

We consider the mass-constrained variational problem for the mixed-dispersion fourth-order Hamiltonian

$$H(u) = \left\|(\Delta + k^2)u\right\|_2^2 - \frac{\beta}{2}\|u\|_4^4 + \frac{\gamma}{3}\|u\|_6^6 \quad \text{on } H^2(\mathbb{R}^2;\mathbb{C}),$$

where $k, \beta, \gamma > 0$. We give a self-contained sufficient condition for strict subadditivity and compactness: if

$$0 < m \le m_B := \frac{3\beta}{\gamma\sqrt{k^4 + 3\beta^2/(32\gamma)}},$$

then the constrained infimum $I(m)$ is negative, strictly subadditive at every positive split, and attained. More precisely, every minimizing sequence possesses a subsequence converging strongly in H^2 after spatial translations. The argument uses anisotropic oscillatory Gaussian trials, an exact mass-preserving dilation identity, an $H^2 \to L^\infty$ estimate, and a concentration–compactness splitting argument. No radial, periodic, or finite-cavity restriction is imposed. The threshold is a sufficient one; its sharpness and uniqueness or symmetry of minimizers are not asserted.

1 Setting and main result

Fix $k, \beta, \gamma > 0$, put $X = H^2(\mathbb{R}^2;\mathbb{C})$, and write

$$A := \Delta + k^2, \qquad M(u) := \|u\|_2^2, \tag{1.1}$$

$$D(u) := \|\Delta u\|_2^2, \qquad K(u) := \|\nabla u\|_2^2, \tag{1.2}$$

$$T(u) := \|Au\|_2^2, \qquad Q(u) := \|u\|_4^4, \tag{1.3}$$

$$S(u) := \|u\|_6^6, \qquad H(u) := T(u) - \frac{\beta}{2}Q(u) + \frac{\gamma}{3}S(u). \tag{1.4}$$

All norms and integrals are taken over $\mathbb{R}^2$. For $m > 0$, define

$$I(m) := \inf\{H(u) : u \in X,\ M(u) = m\}, \qquad I(0) := 0. \tag{1.5}$$

We use the standard inhomogeneous H^2 topology; by Fourier interpolation, $\|u\|_2 + \|\Delta u\|_2$ is an equivalent norm on X.

Theorem 1.1 (Whole-space existence and compactness)**.** *Let*

$$\rho := k^4 + \frac{3\beta^2}{32\gamma}, \qquad m_B := \frac{3\beta}{\gamma\sqrt{\rho}}. \tag{1.6}$$

For every $0 < m \le m_B$ the following hold:

(i) *The infimum is finite and negative:*

$$-\frac{3\beta^2}{16\gamma}m \le I(m) < 0. \tag{1.7}$$

(ii) *For each $0 < \mu < m$, strict binding holds:*

$$I(m) < I(\mu) + I(m-\mu). \tag{1.8}$$

(iii) *If $M(u_n) = m$ and $H(u_n) \to I(m)$, then some subsequence, translations $y_n \in \mathbb{R}^2$, and $u \in X$ satisfy*

$$u_n(\cdot + y_n) \longrightarrow u \quad \text{strongly in } H^2(\mathbb{R}^2), \quad M(u) = m, \quad H(u) = I(m). \tag{1.9}$$

Consequently the minimizing set

$$\mathcal{G}_m := \{u \in X : M(u) = m,\ H(u) = I(m)\}$$

is nonempty and sequentially compact modulo spatial translations.

The proof is assembled in Sections 2–4. The argument does not assert uniqueness or a prescribed symmetry of a minimizer, nor that a specified periodic or numerically computed branch coincides with a whole-space minimizer.

2 Coercivity and strict negativity

Lemma 2.1 (Lower bound and constrained coercivity)**.** *For all $u \in X$,*

$$H(u) \ge T(u) - CM(u), \qquad C = \frac{3\beta^2}{16\gamma}. \tag{2.1}$$

For each fixed $m > 0$, every sequence with $M(u_n) = m$ and $\sup_n H(u_n) < \infty$ is bounded in X.

Proof. The pointwise identity, valid for $r \ge 0$,

$$-\frac{\beta}{2}r^2 + \frac{\gamma}{3}r^3 = \frac{\gamma}{3}r\left(r - \frac{3\beta}{4\gamma}\right)^2 - \frac{3\beta^2}{16\gamma}r \tag{2.2}$$

gives (2.1) upon taking $r = |u|^2$ and integrating. For a fixed mass m, this implies $T(u_n) \le H(u_n) + Cm$, hence $\sup_n T(u_n) < \infty$. Since $\Delta u = Au - k^2 u$,

$$\|\Delta u_n\|_2 \le \|Au_n\|_2 + k^2\sqrt{m}.$$

Finally, Plancherel and Cauchy–Schwarz imply

$$\|\nabla u_n\|_2^2 \le \|u_n\|_2 \, \|\Delta u_n\|_2 .$$

Together these bounds give uniform control of the H^2 norm. □

Lemma 2.2 (Negative energy at every positive mass)**.** *For every $m > 0$, $I(m) < 0$. In particular, (1.7) holds at every positive mass.*

Proof. For $a, b, R > 0$ set

$$f_{a,b}(x,y) = (\pi ab)^{-1/2} \exp\left(-\frac{x^2}{2a^2} - \frac{y^2}{2b^2}\right), \qquad u_R(x,y) = \frac{\sqrt{m}}{R} f_{a,b}(x/R, y/R) e^{ikx}.$$

One checks that $\|f_{a,b}\|_2 = 1$ and $M(u_R) = m$. Writing $g_R = \sqrt{m} R^{-1} f_{a,b}(\cdot/R)$, the real-valuedness of g_R gives

$$A(e^{ikx} g_R) = e^{ikx}(\Delta g_R + 2ik\partial_x g_R), \qquad T(u_R) = \|\Delta g_R\|_2^2 + 4k^2 \|\partial_x g_R\|_2^2 .$$

The Gaussian integrals yield

$$\begin{aligned} \|\partial_x f_{a,b}\|_2^2 &= \frac{1}{2a^2}, \\ \|\Delta f_{a,b}\|_2^2 &= \frac{3}{4a^4} + \frac{1}{2a^2b^2} + \frac{3}{4b^4}, \\ \|f_{a,b}\|_4^4 &= \frac{1}{2\pi ab}, \qquad \|f_{a,b}\|_6^6 = \frac{1}{3\pi^2 a^2 b^2}. \end{aligned}$$

Consequently

$$H(u_R) = \frac{c_2}{R^2} + \frac{c_4}{R^4}, \tag{2.3}$$

where

$$c_2 = \frac{2mk^2}{a^2} - \frac{\beta m^2}{4\pi ab}, \tag{2.4}$$

$$c_4 = m\left(\frac{3}{4a^4} + \frac{1}{2a^2b^2} + \frac{3}{4b^4}\right) + \frac{\gamma m^3}{9\pi^2 a^2 b^2} > 0. \tag{2.5}$$

Choose $a/b > 8\pi k^2/(\beta m)$, which makes $c_2 < 0$. At $R^2 = -2c_4/c_2 > 0$, (2.3) becomes

$$H(u_R) = -\frac{c_2^2}{4c_4} < 0.$$

Since this is an admissible mass-m test function, $I(m) \le H(u_R) < 0$. Combine with Lemma 2.1. □

3 An exact dilation identity and the binding estimate

Lemma 3.1 (Adjustment of arbitrary minimizing sequences)**.** *Let $a > 0$ and let $M(u_n) = a$, $H(u_n) \to I(a)$. There is another mass-a minimizing sequence (w_n), bounded in X, such that every w_n satisfies*

$$D(w_n) - k^2 K(w_n) - \frac{\beta}{4} Q(w_n) + \frac{\gamma}{3} S(w_n) = 0. \tag{3.1}$$

Proof. For $u \in X$ with $M(u) = a$, define the mass-preserving spatial scaling $u_\lambda(x) = \lambda u(\lambda x)$ for $\lambda > 0$. In two dimensions,

$$M(u_\lambda) = M(u), \tag{3.2}$$

$$H(u_\lambda) = \lambda^4 \left(D(u) + \frac{\gamma}{3} S(u)\right) - \lambda^2 \left(2k^2 K(u) + \frac{\beta}{2} Q(u)\right) + k^4 a. \tag{3.3}$$

Put $P = D(u) + \gamma S(u)/3 > 0$ and $B = 2k^2 K(u) + \beta Q(u)/2 > 0$. Here $P, B > 0$ because u has positive mass. As a function of $z = \lambda^2 > 0$, the energy is $Pz^2 - Bz + k^4 a$, with its unique minimum

at $z = B/(2P) > 0$. For each u_n, choose this optimal dilation, denoting the result by w_n. Then $M(w_n) = a$ and

$$I(a) \le H(w_n) \le H(u_n) \longrightarrow I(a).$$

Lemma 2.1 bounds (w_n) in X. Differentiating the explicit scalar polynomial (3.3) at its minimizing scale gives $4(D + \gamma S/3) - 2(2k^2 K + \beta Q/2) = 0$, all quantities now evaluated at w_n. Division by 4 gives (3.1). No differentiability of a spatial scaling orbit in H^2 is required. □

Lemma 3.2 (Uniform quartic–sextic ratio bound)**.** *Let $w \in X$ have mass $a > 0$ and satisfy (3.1). With ρ as in (1.6),*

$$D(w) \le \rho a, \tag{3.4}$$

and

$$\frac{S(w)}{Q(w)} \le \|w\|_\infty^2 \le \frac{\sqrt{\rho}}{4} a. \tag{3.5}$$

Proof. The expansion

$$T(w) = D(w) - 2k^2 K(w) + k^4 a$$

and (3.1) imply

$$D(w) + T(w) = k^4 a + \frac{\beta}{2} Q(w) - \frac{2\gamma}{3} S(w).$$

Completing the square produces the exact identity

$$D(w) + T(w) + \frac{2\gamma}{3} \int_{\mathbb{R}^2} |w|^2 \left(|w|^2 - \frac{3\beta}{8\gamma} \right)^2 \mathrm{d}x = \left(k^4 + \frac{3\beta^2}{32\gamma} \right) a = \rho a. \tag{3.6}$$

All terms on the left are nonnegative, proving (3.4).

For completeness, use the unitary Plancherel normalization of the Fourier transform. Fourier inversion, weighted Cauchy–Schwarz, and polar coordinates give, for every $\eta > 0$,

$$\begin{aligned} \|w\|_\infty^2 &\le (D(w) + \eta a) \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \frac{\mathrm{d}\xi}{|\xi|^4 + \eta} \\ &= \frac{D(w) + \eta a}{8\sqrt{\eta}}, \end{aligned}$$

because the last Fourier integral is $1/(8\sqrt{\eta})$. Since $w \neq 0$ in L^2, one has $D(w) > 0$; otherwise its Fourier transform would be supported on the measure-zero set $\{0\}$. Set $\eta = D(w)/a$ to obtain

$$\|w\|_\infty^2 \le \frac{\sqrt{aD(w)}}{4} \le \frac{\sqrt{\rho}}{4} a. \tag{3.7}$$

Finally $S(w) = \int |w|^6 \le \|w\|_\infty^2 \int |w|^4$, while $Q(w) > 0$, proving (3.5). □

Proposition 3.3 (Strict decrease of energy per mass)**.** *Let $e(a) := I(a)/a$. For $0 < a < b$ with $a + b < 2m_B$, put*

$$h(a, b) := 1 - \frac{a + b}{2m_B} > 0.$$

Then

$$e(a) - e(b) \ge \frac{(b - a)h(a, b)}{a^2} [-I(a)] > 0. \tag{3.8}$$

In particular, e is strictly decreasing on $(0, m_B]$.

Proof. Take the dilation-adjusted mass-a minimizing sequence (w_n) of Lemma 3.1, and set $t = b/a > 1$. Amplitude scaling preserves membership in X and gives $M(\sqrt{t}\, w_n) = b$. Directly,

$$\frac{H(\sqrt{t}\, w_n)}{t} - H(w_n) = (t-1)\left[-\frac{\beta}{2}Q(w_n) + \frac{\gamma(t+1)}{3}S(w_n)\right]. \tag{3.9}$$

By Lemma 3.2, $S(w_n) \le (\sqrt{\rho}/4)aQ(w_n)$. Recalling $m_B = 3\beta/(\gamma\sqrt{\rho})$ and $ta = b$, we have

$$\begin{aligned} -\frac{\beta}{2}Q(w_n) + \frac{\gamma(t+1)}{3}S(w_n) &\le \left[-\frac{\beta}{2} + \frac{\gamma\sqrt{\rho}}{12}(a+b)\right]Q(w_n) \\ &= -\frac{\beta}{2}h(a,b)Q(w_n). \end{aligned}$$

Moreover, $T, S \ge 0$ implies

$$H(w_n) \ge -\frac{\beta}{2}Q(w_n), \qquad Q(w_n) \ge -\frac{2}{\beta}H(w_n).$$

Combining these inequalities with (3.9) yields

$$\begin{aligned} \frac{I(b)}{t} &\le \frac{H(\sqrt{t}\, w_n)}{t} \\ &\le H(w_n) - (t-1)\frac{\beta}{2}h(a,b)Q(w_n) \\ &\le [1 + (t-1)h(a,b)]H(w_n). \end{aligned}$$

Passing to $n \to \infty$ and dividing by a gives

$$e(b) \le \left[1 + \frac{b-a}{a}h(a,b)\right]e(a).$$

Rearrangement establishes (3.8); it is strictly positive by Lemma 2.2 and $h(a,b) > 0$. If $0 < a < b \le m_B$, then $a + b < 2m_B$, so the asserted monotonicity follows. □

Corollary 3.4 (All-split strict subadditivity)**.** *For $0 < m \le m_B$ and $0 < \mu < m$,*

$$I(m) < I(\mu) + I(m-\mu).$$

Proof. Proposition 3.3 gives $e(\mu) > e(m)$ and $e(m-\mu) > e(m)$. Therefore

$$\begin{aligned} I(\mu) + I(m-\mu) &= \mu e(\mu) + (m-\mu)e(m-\mu) \\ &> \mu e(m) + (m-\mu)e(m) = I(m). \end{aligned}$$

□

4 Whole-space concentration–compactness

Lemma 4.1 (No local vanishing)**.** *Let $0 < m \le m_B$, and let (u_n) be a minimizing sequence for $I(m)$. Then*

$$\limsup_{n\to\infty} \sup_{y\in\mathbb{R}^2} \int_{B(y,1)} |u_n|^2 \, \mathrm{d}x > 0. \tag{4.1}$$

Proof. By Lemma 2.1, (u_n) is bounded in H^2. Suppose instead that

$$\sup_{y\in\mathbb{R}^2}\int_{B(y,1)}|u_n|^2\,\mathrm{d}x\to 0.$$

Tile $\mathbb{R}^2$ by unit squares Q_j and let Q_j^* be concentric, slightly enlarged squares with uniformly bounded overlap. The local Gagliardo–Nirenberg inequality on translated fixed-size squares gives

$$\int_{Q_j}|u_n|^4\,\mathrm{d}x\le C\,\|u_n\|^2_{L^2(Q_j^*)}\,\|u_n\|^2_{H^1(Q_j^*)}\,.$$

Since each Q_j^* is covered by a uniformly bounded number of unit balls, our vanishing assumption gives $\sup_j\|u_n\|^2_{L^2(Q_j^*)}\to 0$. Summing over the squares and using bounded overlap, we obtain

$$Q(u_n)\le C\left(\sup_j\|u_n\|^2_{L^2(Q_j^*)}\right)\|u_n\|^2_{H^1(\mathbb{R}^2)}\longrightarrow 0. \tag{4.2}$$

The embedding $H^2(\mathbb{R}^2)\hookrightarrow L^\infty(\mathbb{R}^2)$ and the H^2 bound then imply $S(u_n)\le\|u_n\|^2_\infty\,Q(u_n)\to 0$. In fact, already $T(u_n)\ge 0$ gives

$$H(u_n)\ge-\frac{\beta}{2}Q(u_n)\longrightarrow 0\quad\text{in the sense that }\liminf_n H(u_n)\ge 0.$$

This contradicts $H(u_n)\to I(m)<0$. □

Proof of Theorem 1.1. The lower bound and negativity are Lemmas 2.1 and 2.2. Strict subadditivity is Corollary 3.4. We prove compactness and attainment.

Let (u_n) be any sequence with $M(u_n)=m$ and $H(u_n)\to I(m)$. It is bounded in H^2. By Lemma 4.1, after passing to a subsequence there exist $\delta>0$ and translations $y_n\in\mathbb{R}^2$ such that

$$\int_{B(y_n,1)}|u_n|^2\,\mathrm{d}x\ge\delta.$$

Set $w_n(x)=u_n(x+y_n)$. Translation invariance gives $M(w_n)=m$ and $H(w_n)\to I(m)$. By weak compactness in H^2, local Rellich compactness, and a diagonal subsequence, for some $u\in H^2(\mathbb{R}^2;\mathbb{C})$ we have

$$\begin{aligned} w_n&\rightharpoonup u\quad\text{weakly in } H^2(\mathbb{R}^2),\\ w_n&\to u\quad\text{strongly in } L^p_{\mathrm{loc}}(\mathbb{R}^2)\quad(1\le p<\infty),\\ w_n(x)&\to u(x)\quad\text{for almost every } x. \end{aligned} \tag{4.3}$$

The local L^2 convergence shows $\int_{B(0,1)}|u|^2\,\mathrm{d}x\ge\delta$; hence $u\neq 0$. Weak lower semicontinuity of the L^2 norm gives

$$a:=M(u)\in(0,m]. \tag{4.4}$$

Write $v_n=w_n-u$. The Hilbert-space splitting of L^2 and of Au_n (using the weak H^2 convergence) gives

$$M(v_n)=m-a+o(1), \tag{4.5}$$

$$T(w_n)=T(u)+T(v_n)+o(1). \tag{4.6}$$

Since (w_n) is bounded in H^2 and converges almost everywhere, the Brezis–Lieb lemma [1] applied separately at exponents 4 and 6 yields

$$Q(w_n) = Q(u) + Q(v_n) + o(1), \tag{4.7}$$
$$S(w_n) = S(u) + S(v_n) + o(1). \tag{4.8}$$

Consequently,

$$H(w_n) = H(u) + H(v_n) + o(1). \tag{4.9}$$

We do *not* appeal to weak lower semicontinuity of H; the focusing term has the wrong sign for that shortcut.

We next show $a = m$. Suppose for contradiction that $a < m$. Then $M(v_n) \to m - a > 0$. Define the mass-corrected remainders

$$\widetilde{v}_n := \left(\frac{m-a}{M(v_n)}\right)^{1/2} v_n, \qquad M(\widetilde{v}_n) = m - a.$$

The amplitude multiplier tends to one. Moreover, (v_n) is bounded in H^2, hence $T(v_n)$, $Q(v_n)$ and $S(v_n)$ are uniformly bounded. The explicit polynomial dependence of $H(cv_n)$ on $c > 0$ therefore implies

$$H(\widetilde{v}_n) = H(v_n) + o(1). \tag{4.10}$$

By the definition of $I(m-a)$ and (4.10), $H(v_n) \geq I(m-a) + o(1)$; also $H(u) \geq I(a)$. Together with (4.9) this gives

$$I(m) = \lim_{n\to\infty} H(w_n) \geq I(a) + I(m-a),$$

contradicting Corollary 3.4. Thus $a = m$. This rules out every remainder carrying positive L^2 mass, including spatially separated dichotomy and diffuse positive-mass loss.

Now (4.5) gives

$$\|v_n\|_2 \longrightarrow 0. \tag{4.11}$$

The uniform H^2 bound and Sobolev embedding provide $\sup_n \|v_n\|_\infty < \infty$. Hence

$$Q(v_n) \leq \|v_n\|_\infty^2 \|v_n\|_2^2 \to 0, \qquad S(v_n) \leq \|v_n\|_\infty^4 \|v_n\|_2^2 \to 0. \tag{4.12}$$

From (4.9) and (4.12),

$$I(m) = H(u) + T(v_n) + o(1). \tag{4.13}$$

Because $M(u) = m$, we know $H(u) \geq I(m)$; and $T(v_n) \geq 0$. Therefore (4.13) forces

$$H(u) = I(m), \qquad T(v_n) = \|Av_n\|_2^2 \longrightarrow 0. \tag{4.14}$$

Finally,

$$\|\Delta v_n\|_2 \leq \|Av_n\|_2 + k^2 \|v_n\|_2 \to 0, \qquad \|\nabla v_n\|_2^2 \leq \|v_n\|_2 \|\Delta v_n\|_2 \to 0.$$

Together with (4.11), this proves $v_n \to 0$ strongly in H^2. Thus $w_n \to u$ strongly in H^2 and $M(u) = m$, $H(u) = I(m)$. This proves (1.9) and nonemptiness of $\mathcal{G}_m$. Applying the same argument to an arbitrary sequence in $\mathcal{G}_m$ proves sequential compactness modulo translations. □

5 Frozen-parameter example and limitations

Remark 5.1 (Frozen two-dimensional D4 comparison). For the frozen numerical parameters considered in the source audit,

$$k = 1, \qquad \beta = \gamma = \frac{4608178181141995}{1125899906842624}, \qquad m_{D4} = \frac{1432536998431173}{562949953421312},$$

the quantity ρ is exactly

$$\rho = \frac{49853331562389953}{36028797018963968}.$$

A direct rational check gives

$$9 - \rho m_{D4}^2 = \frac{4547087123240875842124766626783919789007755 91}{1141798154164767904846628775559596109106197299 2} > 0.$$

As $m_B = 3/\sqrt{\rho}$ for $\beta = \gamma$, the check proves $m_{D4} < m_B$ *without relying on rounded decimals.* Numerically,

$$m_{D4} \approx 2.54469689485712, \qquad m_B \approx 2.55034563635155.$$

Here "D4" labels the source problem's square-symmetry setting in *two* spatial dimensions, not a statement about four spatial dimensions.

Remark 5.2 (Scope). The theorem establishes existence of at least one whole-space ground state and compactness of the complete minimizing set modulo translations. It does not establish uniqueness, radial or square symmetry, identification with a particular finite-torus numerical profile, a parent higher-derivative field-theory derivation, or a prediction about higher-dimensional confinement lifetimes. The threshold m_B is sufficient, not claimed sharp.

Remark 5.3 (Relation to earlier fourth-order results). The earlier mixed-dispersion variational problem with a single nonlinear power has substantial predecessors, including Fernández, Jeanjean, Mandel, and Mariş [2]. The present argument addresses the stated combined focusing-cubic and defocusing-quintic Hamiltonian and its sufficient mass threshold. This is not a claim of literature-wide novelty.

References

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[3] R. J. Reyes, *Geometry of Resonance*, proof notes: https://github.com/rickyjreyes/geometry_of_resonance/tree/main/docs/d4_binding.